

REVIEW

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Cybersecurity challenges and solutions in 5G-enabled internet of things

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Abstract

Due to the growing interconnection and sensitive nature of IoT applications, this article provides a comprehensive overview of cybersecurity in 5G-enabled Internet of Things (IoT) networks. The review highlights the significant potential of 5G technology while also emphasizing the need for ongoing work to mitigate its weaknesses and fully harness its advantages. In this context, prospective research avenues are crucial to enhance the resilience and security of 5G-enabled systems. The future directions outlined in this document are diverse and seek to tackle significant deficiencies in 5G-IoT cybersecurity research. A primary objective is to align theoretical analyses with practical implementations by examining specific assaults in IoT contexts, thus improving the comprehension of real-world vulnerabilities. Furthermore, as the 5G standard progresses through its implementation, continuous research must remain synchronized with improvements, focusing on components and features that are currently in development. Formulating effective protection and mitigation strategies to address sophisticated threats is essential for safeguarding this integrated environment. Furthermore, enhancing standards and regulatory frameworks is essential to ensure uniform security procedures across various businesses and areas, especially considering the global nature of 5G-IoT applications. Ultimately, investigating sophisticated attack scenarios that leverage the interaction between IoT devices and 5G infrastructure would enhance understanding of vulnerabilities and aid in the development of more resilient security measures. These directives underscore the necessity for ongoing, flexible research to guarantee the robustness and safety of 5G-enabled IoT networks.

Keywords 5G Networks, Internet of things (IoT), Cybersecurity, 5G-IoT security, Intrusion detection systems (IDS), Federated learning, Machine learning-based security, AI-driven threat detection

1 Introduction

The IoT is a rapidly expanding system of interconnected computing devices that can receive and send data using predefined protocols [1, 2]. According to authors [3], 34 billion IoT devices are now deployed across diverse domains, creating significant challenges for data management due to the growth in quantity, diversity, velocity, and quality



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of data. According to [3], the IoT ecosystems are now dependent on the launch of 5G mobile networks. With its ultra-low latency (less than 2 milliseconds), high throughput, and energy-efficient connectivity, the 5G infrastructure allows for the connection of several devices with different communication and processing needs [4–8]. Integrating 5G with the IoT has led to the development of new access and transmission technologies, including full duplex (FD) communications, millimeter wave (mmWave) frequencies, and machine-to-machine (M2M) communication frameworks [9–16]. When combined, these technologies simplify the scalability and speed of providing sophisticated IoT services [17–20]. Expanding upon previous 4G LTE networks, 5G greatly improves capacity, latency, and reliability [21]. In addition to customizable duplexing, 5G networks offer gigabit-level capacity. They are capable of supporting up to 65,000 concurrent connections and can transfer data at speeds up to 20 times faster than 4G. These characteristics are especially important for enabling low-latency Internet of Things applications, which are crucial in fields like smart homes, autonomous vehicles, telemedicine, and industrial automation [22–26]. Some technical and security problems still exist, even with these upgrades. Problems with optimising bandwidth, devices that consume a lot of power, and keeping connected over long distances are discussed by Uslu et al. [27]. The authors [28] also address the issue of smart devices enabled by 5G having higher power consumption and battery needs. Moreover, problems with coverage and dependability are common with widely used sensors, especially in contexts that are continually changing or have limited resources [29]. In order to tackle the scalability and flexibility problems in the IoT, some have suggested using software-defined networking (SDN) [30–32]. The need to employ smart tactics for managing networks is demonstrated here.

While several studies have looked at different parts of 5G-IoT integration, not many have tackled the standards and technological requirements for long-term interoperability, security, and sustainable growth [21, 33–35]. Although they did not go into much detail about new security vulnerabilities, the authors [33] highlighted connectivity, traffic volatility, and quality of service assurance as major concerns. Similarly, Alsibahee et al. [35] examined methods for data protection and trust establishment; however, they did not thoroughly assess the efficacy of existing security frameworks or their limitations. The authors [34] examined the advantages and disadvantages of 5G network slicing for IoT applications in a separate publication. They discussed both its potential and the associated risks. The proposed solutions primarily consisted of concepts and lacked substantial details on their implementation.

The majority of the recent work concentrated on privacy, security, or architectural facilitators. These papers [26, 33, 35–42] examine the critical requirements, privacy-preserving methodologies, and technical facilitators within the layered architecture of 5G-enabled IoT systems. They provides scholars, professionals, and policymakers with critical insights to facilitate the development of secure, robust, and scalable IoT infrastructures compatible with 5G technology.

1.1 Review of existing surveys

Table 1 presents the primary subjects of recent review studies that show a comprehensive qualitative comparison between our survey and other significant studies in the domain of 5G-enabled IoT security. Our study covers a wide range of risks, attack vectors, mitigation strategies, and architectural considerations, which stands out for

Table 1 Qualitative comparison of our survey with related work on 5G-enabled IoT security

Study	Year	5G-IoT architecture	Security focus	Technical enablers	Challenges discussed	Novel contributions
<i>This proposed survey (ours)</i>	2025	Comprehensive analysis of layered 5G-IoT architecture	Broad coverage of threats, attacks, and defense mechanisms	Covers FL, AI/ML, edge/fog, blockchain, SDN/NFV	Explores scalability, heterogeneity, latency, privacy, and 6G evolution	Presents a multi-perspective taxonomy with forward-looking insights
Wazid et al. [43]	2020	General overview of 5G and IoT elements	Focus on common security threats and risk management	Discusses ML and cryptographic primitives at a high level	Touches on integration and standardization issues	Lacks architectural breakdown or taxonomy
Scott-Hayward et al. [44]	2016	Descriptive architectural model (Focused on SDN)	Emphasizes threat models and countermeasures (in SDN/NFV)	Highlights AI and SDN with examples	Covers privacy and device-level vulnerabilities (in SDN/NFV)	No unique classification framework or 6G outlook
Stellios et al. [45]	2018	Focused on critical infrastructure (pre-5G)	Limited security scope (device-level vulnerabilities and attack paths)	Mentions IDS and authentication methods	Little discussion of system-level challenges	No focus on emerging paradigms or layered architecture
Meneghillo et al. [46]	2019	General IoT integration overview (Focus on threats)	Discusses protocol-level threats and mitigation	Introduces blockchain and MEC briefly	Identifies latency and interoperability issues	Descriptive but lacks detailed taxonomy or evaluation
Heidari et al. [48]	2023	Partial architectural coverage (Intrusion Detection)	Focused on intrusion detection systems	Reviews machine learning methods for IDS	Mentions data integrity and trust challenges	No survey-level classification or framework
Alcaraz et al. [47]	2023	Discusses Digital Twin architecture	Reviews security risks in 6G/Digital Twin environments	Covers AI/ML and resilience	Addresses data security and reliability	Narrow focus on Industry 5.0 and Digital Twins
Nguyen et al. [3]	2021	Brief architectural reference (6G/IoT)	Strong emphasis on key technological enablers	Lacks discussion on enablers like FL or SDN	Discusses key security challenges in IoT-6G	Provides attack taxonomy but no technical synthesis
Singh Kaunert [30]	2024	Focused on health monitoring legal aspects	General vulnerabilities view	Explores 5G connection and legal/ethical guidance	No taxonomy; focuses on a single aspect of 5G cybersecurity	Systematic legal analysis with futuristic outcomes
Sefati et al. [49]	2025	QoS-driven IoT architectural analysis	Evaluates attack detection methods (signature, anomaly, hybrid) and their QoS maintenance under DDoS, MITM, data breaches	AI/ML-driven anomaly detection; hybrid detection; QoS-centric evaluation	Resource constraints; QoS degradation under attacks; lack of adaptive recovery models	Provides a QoS-centered survey with cross-domain insights for 5G-enabled IoT

its thorough and multi-faceted architectural investigation of 5G-IoT systems. Modern technologies such as Federated Learning (FL), artificial intelligence (AI), machine learning (ML), blockchain, edge computing, fog computing, and software-defined networking (SDN/NFV) are all part of it.

Our research thoroughly investigates crucial subjects such as scalability, heterogeneity, latency limitations, privacy protections, and the future of 6G networks, in contrast to previous surveys that usually focus on specifics such as threat taxonomies or device-level security or general overviews without discussing architectural details. Our unique

contribution: a multifaceted taxonomy that integrates previous work and suggests new avenues for inquiry.

Although earlier studies by authors [43, 44] provide a basic outline of 5G and IoT security, these studies do not address the issues of next-generation network compatibility and detailed architectural modelling. Prior research has focused on out-of-date IoT systems [45], narrowly defined technology enablers like blockchain and MEC [46], or domain-specific applications like smart cities [47]. Although taxonomies of attacks and threats have been discussed in certain articles [3], there is a lack of integration with technical enablers and architectural considerations at the system level.

Moreover, recent research articles have been considered for an IoT cybersecurity review using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach in [30]. The authors focused on IoT risks in 5G network components, such as intermediate network and end devices. However, the authors made a great contribution, but they shortened the taxonomy and evaluation.

Our survey is compared to similar research in terms of methodology, taxonomic depth, scope, and gaps in knowledge in the comparative analysis. It looks like this is the first survey of its kind to combine a taxonomy of layers with an analysis of cybersecurity in 5G-enabled Internet of Things systems, taking resilience and standards into account. In addition, the planned poll shows how comprehensive and deep our study is, drawing attention to its practical applications and implications for those working to build 5G-enabled Internet of Things infrastructures that are safe, scalable, and future-proof.

1.2 Contributions

Research on 5G and beyond-enabled IoT security and resilience already exists, as we indicated earlier. Nevertheless, current SLRs and surveys do not adequately cover the topic of cybersecurity and resilience as it pertains to Internet of Things applications enabled by 5G and future generations. Further, the current literature is vague when it comes to the precise strategies utilised to ensure the IoT is secure and resilient when 5G and later networks are operational. Therefore, by delving into the subjects discussed in Sect. 3, our research aims to address these deficiencies. The main results of the study are:

- We provide a comprehensive and current overview of existing studies that have focused on cybersecurity and resilience in IoT ecosystems that have been improved by 5G and others.
- We list the best practices for next-generation IoT network security, including architectural frameworks, performance metrics, threat models, and validation processes.
- We provide the issues related to scalability, heterogeneity, latency, interoperability, privacy, trust management, and adversarial threats are included in the proposed multi-faceted framework for classifying cybersecurity in 5G-IoT.
- We provide possible solutions and approaches to lessen the effects of new technology developments, including federated learning, blockchain, anomaly detection using artificial intelligence and machine learning, software-defined networking (SDN), and network function virtualisation (NFV).
- We analyse the resilience needs of 5G-IoT networks, with a focus on fault tolerance, adaptive security, and self-healing for applications that are mission-critical and run in real-time.

- We delve into how network slicing, edge/fog computing, Multi-Access Edge Computing (MEC), and other enabling technologies could improve the safety and efficiency of distributed IoT networks.
- We detail the current benchmarking and standardisation initiatives, drawing attention to the fact that they lack practical application, experimental validation, and reproducibility.
- Finally, we outline a plan for future 6G deployment and a set of guidelines for designing secure, scalable IoT systems.

2 Taxonomy

The cybersecurity in 5G-enabled IoT systems is divided into four main groups by this taxonomy, as shown in Fig. 1: 5G enablers, IoT components, security needs, and security objectives. Each domain addresses critical components that establish the safety of these interdependent ecosystems; together, they provide a comprehensive framework for comprehending their complexity and guiding future research directions.

2.1 5G facilitators

The IoT ecosystems of the future will consist of 5G networks as their foundation. These are brand new and have features like increased capacity, vast device connectivity, and extremely low latency. New opportunities for service delivery and performance improvement are presented by these technological enablers, but the cybersecurity attack surface is also increased. The primary 5G enablers are examined in this part along with their functions and the security flaws that must be addressed in order to ensure the safety of 5G-enabled IoT systems.

2.1.1 Network segmentation

Network slicing allows for the virtualisation and separation of a physical 5G network into multiple independent network instances, each optimised to meet specific service needs and quality-of-service (QoS) criteria [50, 51]. The ability to run several tenants on

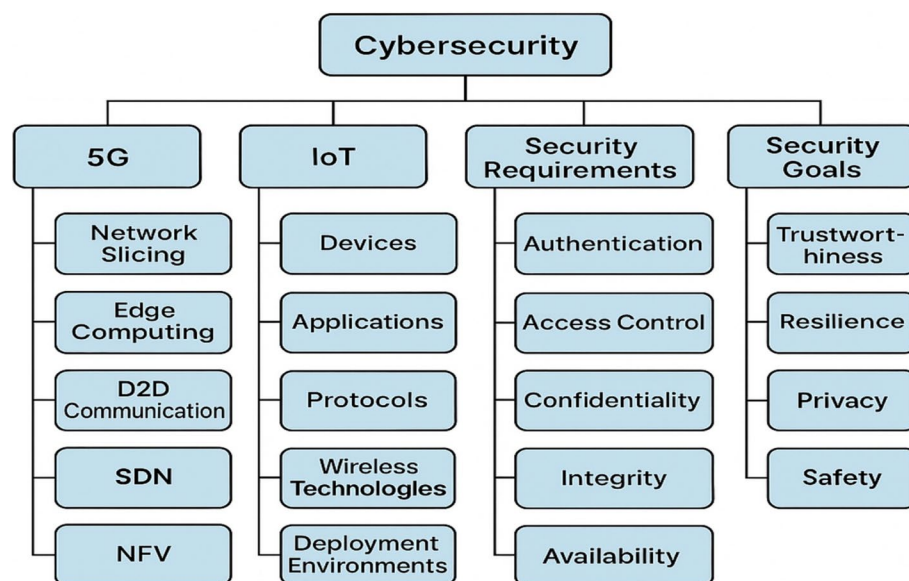


Fig. 1 Taxonomy for cybersecurity in 5G-enabled IoT

the same infrastructure is essential for the IoT to support a wide variety of uses, such as healthcare monitoring and industrial automation. In order to stop illegal data leakage or lateral movement across slices, network slicing requires tight isolation. All services' privacy and security may be at risk if cross-slice assaults were to happen if slice isolation were lost. The security guarantees included in service-level agreements (SLAs) must be followed by slices. Constant vigilance and ever-changing policy enforcement are needed for maintaining slice-specific security measures [16].

2.1.2 Edge computing

With edge computing, analytics and data processing are moved closer to where the Internet of Things data originates. According to the edge computing survey, this allows people to make decisions in real-time by reducing latency and bandwidth consumption. This decentralised design has many benefits, but it also makes security more difficult. Due to the different contexts and their accessibility, establishing trust at the perimeter is tough. Due to their low processing capability, edge nodes frequently necessitate efficient and robust authentication and encryption methods [52]. In order to prevent eavesdropping and manipulation of computations transmitted to edge nodes or between edges and the cloud, secure task offloading techniques are crucial. In order to protect themselves from cyber and physical dangers, edge devices need stronger hardware and software protections [10].

2.1.3 Device-to-device (D2D) communication

Devices connected to the IoT can avoid middlemen and talk to each other directly through device-to-device (D2D) connection. The authors of the cited work state that this improves spectrum usage and speeds up communication. Both automotive networks and emergency response systems benefit from this type of peer-to-peer technology because of the importance of quick local coordination in these contexts. The fact that the network perimeter is exposed to multiple potentially vulnerable endpoints makes direct-to-device communication a major security risk for networks. According to the authors [53], proximity-based attacks like spoofing, jamming, and unlawful interception become easier to execute when there is no controlling authority. Mutual authentication, real-time intrusion detection, and safe key exchange should be capabilities of security frameworks developed for direct-to-device (D2D) systems [15].

2.1.4 Software-defined networking (SDN)

Network administration is transformed by SDN's separation of the data plane and the control plane. Programmatic central administration of network flows becomes possible as a result of this [44]. The ability to swiftly reconfigure networks and apply policies in response to new threats or changing circumstances is a direct result of its adaptability. The SDN controller becomes both an attractive target and a potential vulnerability when the control plane is centralised, which poses a significant security risk [14]. A controller hack might cause disruptions to the network or unauthorised alterations to the flow of traffic. Implementing SDN requires strong security measures to avoid denial-of-service attacks and controller compromise. These methods include encrypted communication channels, redundant controllers, and powerful access controls [54].

2.1.5 Network function virtualisation (NFV)

According to the NFV overview [55], traditional network tasks like firewalls, load balancers, and intrusion detection systems are turned into software instances that run on shared physical hardware. While this virtualization improves scalability and flexibility, it does so at the expense of security. It is more likely to occur in a shared hardware environment when a hostile virtual machine (VM) takes control of the underlying hypervisor or other VMs that are co-located in hypervisor escape attacks [56, 57]. Data integrity or functionality can be compromised if virtual computers are not kept segregated. In addition, misconfiguration or resource contention could compromise security. Hypervisor fortification, virtual machine isolation, secure lifecycle management, and continuous monitoring are all essential virtualisation security measures for network function virtualisation (NFV) settings.

To solve these problems, there is a need for new trust frameworks that can balance the demand for real-time processing with the limitations imposed by device resources and strong security. Quick and robust endpoint authentication approaches that preferably use hardware-rooted security features are preferable. To safeguard 5G-enabled IoT environments from the ever-increasing attack surface, adaptive threat mitigation, constant security monitoring, and dynamic, context-aware access control policies are crucial.

To conclude, 5G paves the way for game-changing Internet of Things applications; however, due to security flaws, it is imperative to implement all-encompassing defenses at the same time, including architectural, operational, and technical controls, to ensure the availability, confidentiality, integrity, and reliability of the system.

2.2 Internet of things components

Hardware, software, protocols, wireless networks, and infrastructure for deploying IoT devices make up an ecosystem for the IoT. Because of the specific security risks posed by each type, an all-encompassing, multi-pronged, and context-aware approach is required.

2.2.1 Devices

IoT devices such as sensors, actuators, and mobile nodes don't always have enough power, memory, or computing for their intended use. Limitations that prevent the deployment of current security hardware, like Hardware Security Modules (HSMs) or Trusted Platform Modules (TPMs), lead to an inadequate root-of-trust infrastructure [58–60]. The security vulnerabilities of devices are increased due to their susceptibility to manipulation, side-channel assaults, and hardware-level invasions, as they can be physically accessible [61–67]. Furthermore, due to a lack of safe boot techniques and hardware-enforced isolation, numerous IoT devices have firmware vulnerabilities. With so many vendors and device designs out there, it can be challenging to ensure secure lifecycle management, which includes safe provisioning, upgrades, and decommissioning [68–73].

2.2.2 Applications

Many fields could benefit from Internet of Things applications; smart transportation, healthcare, industrial automation, and environmental monitoring are just a few examples. Industries have different security and privacy needs due to the varied kinds and amounts of sensitive data they handle. protecting applications that utilise the Internet

of Things [74–77]. In compliance with laws like HIPAA, healthcare applications must firmly safeguard patients' PHI. In contrast, the primary goal of IIoT applications is to ensure that operations remain online and uninterrupted at all times. Combining older and modern systems makes interoperability even more of a challenge because of the wide variety of application environments. Difficulty is in creating access control policies and enabling secure data movement across several platforms. A security architecture that can adapt to different situations and procedures is therefore essential [17, 78–80].

2.2.3 Protocols for communication

The ease of use and low resource consumption of lightweight communication protocols such as MQTT, CoAP, and 6LoWPAN make them popular among IoT devices [17]. Security is frequently neglected in the name of efficiency and user-friendliness by these protocols. Due to a lack of built-in support for end-to-end encryption and robust authentication, session hijacking, message interception, and forging were all feasible in the early days of MQTT. Datagram Transport Layer Security (DTLS) is used by CoAP, but there are still issues with key management and threats of resource exhaustion. Additionally, due to the lack of standardisation in application protocols for securely initiating and configuring connections, IoT deployments are susceptible to replay attacks and persistent man-in-the-middle (MITM) attacks [81–84].

2.2.4 Wireless technologies

Wireless communication technologies such as LoRaWAN, NB-IoT, and ZigBee form the backbone of the IoT infrastructure. They make it possible to transmit data over different distances and power distributions. However, wireless channels can be targeted by many types of attacks, such as jamming, eavesdropping, spoofing, and replay assaults due to their broadcast medium nature [85–88]. There is a risk of replay and message injection attacks on LoRaWAN networks due to insufficient cryptographic key protection. While NB-IoT has the safety features of cellular networks, it also takes on the challenges and weak spots of mobile networks. There are additional hazards associated with ZigBee's ability to construct mesh networks, such as routing attacks and compromised nodes spreading misleading information. A number of measures, such as frequency hopping, lightweight encryption, anomaly detection, and secure key distribution protocols, are required to guarantee the availability, privacy, and security of wireless IoT communications [89–91].

2.2.5 Deployment environments

From smart homes and offices to smart factories and vast smart city infrastructures, there are countless possible uses for the Internet of Things. Problems with physical safety and trust boundaries are unique to each setting. A lack of strong security measures makes it more likely that smart home gadgets may be physically tampered with or that unauthorised individuals could gain access to the local network [92–94]. On the flip side, strict safety and cybersecurity rules need to be followed by industrial installations to avoid putting people at risk or interrupting operations due to breaches. Complex trust issues among businesses and difficulties with regulatory compliance are inherent in urban-scale IoT networks. Therefore, it is critical to create deployment-specific threat

models and adapt security controls, such as network segmentation, physical security, and secure identity and access management, to the unique circumstances [95–97].

2.2.6 Critical analysis

The security landscape is fragmented and difficult due to the extremely diversified IoT ecosystem. Universal security frameworks are difficult to create due to the wide variety of device types, protocol standards, and deployment conditions. Inability to do over-the-air (OTA) firmware upgrades is a problem with many older and cheaper IoT devices, which makes it harder to fix them quickly and leaves them vulnerable to long-term security threats [1]. Implementing end-to-end security requirements in systems with several vendors can be challenging due to proprietary communication protocols that are not always interoperable. Trust is severely damaged, and device integrity and compliance verification are made more difficult by the lack of standardised security certification.

Security solutions need to use scalable, adaptive, and interoperable architectures to solve these problems. To ensure efficient use of resources, light cryptographic algorithms, hardware-accelerated security modules, and integrated security management frameworks must function together. Furthermore, adaptive defences for evolving IoT environments may be possible with real-time threat detection and anomaly evaluation utilising contextual security protocols powered by AI. The integrity, availability, and privacy of data in the IoT ecosystem depend on our fully resolving these issues [79].

2.3 Security requirements

Every 5G IoT system must adhere to fundamental technical security standards to lessen risks and vulnerabilities. These features are essential for 5G-IoT deployments to keep trust, privacy, and functionality intact in very diverse and complicated situations.

Authentication is the backbone of secure access control. Identity verification must be performed prior to granting access to the system to any individual or technology. Since IoT networks consist of a diverse array of devices, some of which may be weaker than others in terms of resources, conventional authentication methods do not necessarily apply. Modern approaches that integrate hardware-assisted mechanisms such as Trusted Platform Modules (TPMs) and Physically Unclonable Functions (PUFs) with lightweight cryptographic protocols appear to make quick and secure authentication a reality [79].

Individuals' eligibility to use resources or services is determined by access control systems according to preset roles, policies, or contextual criteria. Adaptable and highly customised access control frameworks are required for 5G-IoT environments due to their dynamic and scalable nature. According to authors [81], these frameworks need to be able to handle a large number of devices and users while also meeting privacy and legal requirements. Access control methods that are attribute and role-based, with the addition of continuous monitoring, are becoming more popular as a means for enterprises to follow security regulations.

Limiting access to sensitive information to authorised individuals only, confidentiality helps avoid leaks and unauthorised disclosures. It is of the utmost importance in IoT settings where sensitive information like personal location data, commercial secrets, or health records is at stake. Cryptographic methods, including end-to-end encryption and efficient algorithms made for low-resource devices, are necessary to safeguard data privacy in 5G-enabled IoT networks.

The data must not change from its static or transitive state until an approved activity modifies it. When it comes to areas like healthcare monitoring and industrial automation, where patient safety is paramount, manipulating or adding false data into integrity-based systems can have serious effects. Data integrity is often guaranteed by means of cryptographic hash functions, digital signatures, and secure logging systems [81, 98, 99].

The timely distribution of resources and services delivered by a network must be uninterrupted and error-free. The effective operation of IoT applications depends on this. New attack vectors for DoS and DDoS-related attacks have surfaced with the introduction of 5G networks and edge computing infrastructures, which allow for the processing of data in real-time. If you want to keep availability high in case of any failures, you need an effective backup plan, intrusion detection systems, and a well-designed network [81].

2.4 Security goals

In addition to technical fixes, security goals can include system-level ones. They explain the overall plan for protecting IoT environments with 5G technology and what changes can be expected down the road. This is our goal as we plan, develop, and oversee our Internet of Things solutions. Security is their top priority throughout the whole system lifecycle, from design to decommissioning.

System trustworthiness is defined as the degree to which users, operators, and regulatory agencies have full faith in the system's ability to securely operate and process data while maintaining its accuracy, integrity, and veracity. In addition to technical robustness, there are other aspects to consider while considering reliability. Acting honestly, responsibly, and morally is a part of it. Automated decision-making is becoming more commonplace, so it is crucial to set up systems that continuously check system functionality and regulatory compliance using audit trails and explainable AI methodologies [100].

A robust system can keep running even when attacked, has a hardware failure, or has network outages. Resilience aims to change the focus from prevention, which is typical of traditional security goals. Rather, what matters most are detection, response, and recovery. An adversarial model, anomaly detector, intrusion recovery system, and fault-tolerant design are all part of a resilient IoT-5G infrastructure. Zero-trust designs that investigate all interactions regardless of their network location can enhance system resilience, according to Resilience IoT and Zero-Trust Architecture. Ensuring the highest attention and respect for individual rights is crucial when it comes to managing personal or sensitive information received, stored, and disseminated within IoT networks. This is a legal and ethical requirement. Different areas have different needs when it comes to privacy. Due to the sensitive nature of patient records and the need to adhere to laws like HIPAA and GDPR, smart healthcare systems necessitate stringent security measures. On the flip side, smart lighting systems might not necessitate as many privacy precautions. In order to find a middle ground between data utility and privacy, privacy-preserving methods such as federated learning, secure multi-party computation, and differential privacy are becoming more important [1].

The goal of safety in IoT networks is to protect users from damage in the event that cyber-physical systems fail or are breached. Security breaches might have serious, real-world repercussions because 5G-IoT settings are often linked to vital infrastructure, industrial control systems, and medical devices. Strict compliance with industry-specific

safety requirements, extensive risk evaluation, ongoing monitoring, and redundant systems are all essential for safety. To prevent security breaches from becoming actual physical dangers, cybersecurity and safety engineering must work together [83].

3 Overview of the 5G-IoT architecture

The shortcomings of 4G networks, upon which the IoT has long depended, are becoming apparent in light of the increasing complexity of IoT applications and the global population. The present IoT is being greatly expanded through the usage of 5G networks, which improve communications services. See Fig. 2 for an illustration of how a 5G-enabled IoT infrastructure, supported by edge computing and smart sensing devices, has ushered in a new era in data transmission and processing. Here, 5G networks allow for near-instantaneous data transfer with nearly no latency [83], allowing for problem-free communication among several smart devices. Edge computing, which is complementary to this configuration, enables processing and analysis of data at the network's perimeter, decreasing reliance on centralised cloud computing and increasing the likelihood of making choices rapidly. On the other hand, state-of-the-art sensors and AI algorithms allow smart sensing devices to accurately gather and evaluate data from the real world. From healthcare and industrial automation to autonomous vehicles and smart cities, 5G wireless connectivity, edge computing, and smart sensing devices have the potential to change a multitude of industries. As a result, we could experience previously unseen degrees of connectedness and efficiency, and our relationship with the world around us would be forever altered. The security and network advantages of the IoT are driving the expansion of the Internet [101]. Among the many issues they introduce are new protocols, security concerns, and large numbers of nodes. 5G enables innovative and efficient solutions to these issues by facilitating the incorporation of new technologies into the IoT. For instance, with 5G-powered IoT, farmers can optimize and track crop conditions in real-time, which might lead to better utilization of resources and higher yields [101].

Modern 5G-enabled IoT systems require more effective and expandable network administration tools. A service-oriented paradigm for network management based on a Service Aggregation and Caching (SAaC) framework was established [83] to meet these objectives. In order to facilitate more nimble administration and communication,

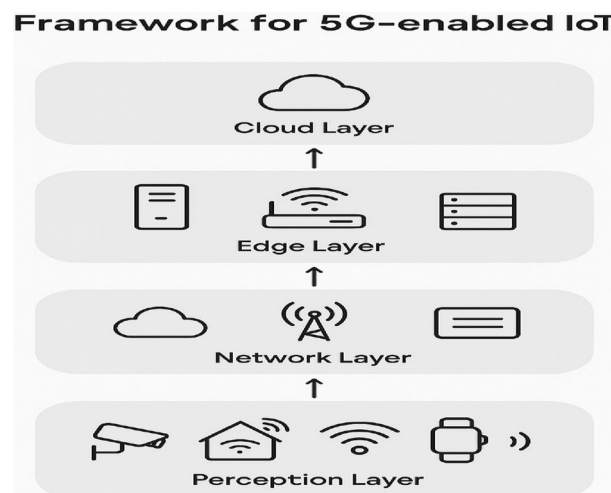


Fig. 2 Framework for 5G-enabled IoT

this paradigm departs from the conventional data-centric architecture by transforming data streams into service modules. With gains like a 20.52–56.09% drop in system reaction time, a 10.85–36.7% reduction in data traffic, and nearly 50% savings in energy consumption, the SAaC mechanism greatly improves network efficiency. These results demonstrate the promise of service aggregation in reducing power requirements and congestion in dense IoT deployments.

On the other hand, optimising energy usage in smart city applications is now a top priority. [83] showed that intelligent control systems, like smart street lighting and environmental sensing, made possible by 5G-enabled IoT systems, can greatly improve sustainability. Both operational efficiency and the larger objective of decreasing carbon emissions and energy footprints are aided by these applications. All things considered, these studies show that smart service design and architectural advances are key to making future IoT infrastructures more efficient and longer-lasting.

A new paradigm for service-oriented network management was established [102] to assist in the effective management of 5G-enabled IoT systems. The SAaC (service aggregation and caching) approach streamlines network administration and reduces network traffic in this design. Creating services out of data is the starting point for SAaC to succeed in a data-centric system architecture. As compared to conventional methods, SAaC methodologies reduced reaction time by 20.52–56.09%, traffic by 10.85–36.7%, and energy consumption by less than 50%. Reduced traffic and energy consumption are two benefits of integrating the services. Using 5G-enabled IoT to intelligently regulate street lighting and environmental monitoring, smart cities can reduce energy consumption and carbon emissions.

4 Wireless network security evaluation

While earlier generations of networks were more secure, 4G networks allow for machine-to-machine (M2M) connectivity and network integration via the IoT, which opens the door to new vulnerabilities and makes critical network servers easily accessible to attackers [102]. While cross-infrastructure assaults were prevalent in 3G and 4G networks, single-infrastructure attacks were more common in 1 G and 2G networks. Eavesdropping, brute force attacks, and network congestion attacks were the main security threats for 1G, 2G, and 3G, respectively. Various security measures have been implemented throughout the shift from 1G to 4G [102]. The increased interconnection of 4G networks with the internet and the IoT introduced more security risks compared to 3G. 4G has several important challenges, including insufficient protection in network control, unprotected radio communication, limits in security design, and user privacy [102]. New security vulnerabilities and threats will emerge as we move to 5G and beyond. This calls for revised approaches and regulations to guarantee safety. Because their security rules have not been updated to reflect new emerging technology, 63% of organisations using IoT services are facing increased security difficulties [102].

4.1 Challenges of 5G: issues of technology and safety

5G, the next big thing in communication tech, will make the virtual world even more linked. With its superior protocol, speed, and network architecture, 5G mobile communication surpasses 4G in every way. Software-Defined Networking (SDN) and Network Function Virtualisation (NFV) are introduced to ensure consistent user experiences,

improve connectivity, increase speed and throughput, decrease latency, and enhance bandwidth [68]. In addition, 5G networks allow for convergent heterogeneous networks, which give customers access to more capacity by integrating licensed and unlicensed wireless technologies.

Regardless of these advantages, there are new security concerns and complexities that come with using SDN and NFV in different situations. Strong security measures are required since 5G will connect communication networks with a plethora of real-time applications. A review of the literature has uncovered the main obstacles to 5G security [68]. The summary of the main security concerns is in Table 2.

- **Security for the User Plane:** Ensuring secure communication between 5G devices and networks is the responsibility of the user plane, which is tasked with securing end-user data.
- **Warning of Storms:** Flash network traffic and signaling storms induced by hacked equipment are more likely in highly connected situations where the number of devices linked to 5G networks is increasing. This poses a substantial danger to the 5G signaling network.
- **Ensuring Safe Mobility:** To ensure the safety of their customers and the integrity of their 5G networks, mobile carriers must establish secure roaming.
- **Blockchain Security:** To guarantee decentralized security and dependable communication channels, 5G blockchain-based communication requires more traffic routing points.
- **Denial-of-Service (DoS) Attacks:** Attacks like these might cause infrastructures or end-user devices to become inaccessible by overwhelming them with traffic.
- **Able to Work With:** 5G devices may put a strain on current security monitoring solutions due to their greater bandwidth requirements.
- **IoT Device Security:** Due to a lack of built-in security, many IoT devices necessitate external evaluation.

Table 2 Summary of key 5G security challenges

Security concern	Description
User plane security	Protection of user data in transit to ensure secure communication between 5G devices and networks
Signaling storms	Risk of flash network traffic and signaling overload caused by compromised or numerous connected devices
Secure roaming	Ensuring safe connectivity and data exchange for users moving across different 5G networks
Blockchain security	Need for decentralized security points and reliable routing in 5G blockchain-based communication
Denial of service (DoS) Attacks	Traffic flooding attacks that disrupt availability of devices and network infrastructure
Compatibility strain	Existing security monitoring tools may struggle with increased bandwidth and device diversity
IoT device security	Many IoT devices lack inherent security, making them vulnerable entry points
Encryption vulnerabilities	Lack of encryption in early connection phases leaves device data exposed
NFV security issues	Include virtualisation threats, API vulnerabilities, management orchestration flaws, and supply chain risks
Layered network threats	Security risks across core network, access network, multi-edge computing, virtualization, physical infrastructure, and general threats like DoS
SDN-related risks	Foundation of 5G network architecture brings inherited security challenges from software-defined networking

- **Vulnerabilities in Encryption:** Data stored on 5G devices is susceptible to assaults since early connection processes are frequently unencrypted.
- **Issues with NFV Security:** Access control and secure communication between virtual and physical systems are concerns raised by open-source software and virtualization in NFVs. Focus areas encompass:
 - Possible dangers associated with virtualization and containerization include attacks on the control plane, API vulnerabilities, assaults on forwarding devices, and fake traffic flows. Inconsistent management and orchestration can lead to protocol vulnerabilities, MANO single points of failure, compromised orchestration, and policy violations.
 - The use of commercial off-the-shelf hardware and open-source software introduces performance and security issues. The use of incorrectly configured APIs exposes important parameters. There are challenges with interoperability between virtual and physical environments.
 - Threats to the supply chain encompass things like viruses, fake components, poor design, and insufficient upkeep.
 - Shielding legitimate interception (LI) operations from other non-fiduciary value (NFV) operations.

5G security risks can be categorized by layers:

- **Core Network Threats:** Including eavesdropping, interception, hijacking, and nefarious activity or abuse (NAA), associated with SDN, NVE, NS, and MANO.
- **Access Network Threats:** Related to EIH affecting radio access technology, radio access networks, and non-3GPP compliant access technologies.
- **Multi-edge Computing Risks:** Targeting components at the network edge, overlapping with NAA and EIH.
- **Virtualization Risks:** Pertaining to NFV and VNF processes and operations.
- **Physical Infrastructure Threats:** Including attacks, equipment damage or loss, service interruptions, and natural disasters impacting network IT infrastructure.
- **Generalized Risks:** Such as DoS attacks, affecting any ICT-enabled network or system.

Finally, concerns about SDN arise because it forms the foundation of 5G, thereby inheriting and amplifying security challenges inherent to software-defined networking. A summary comparison between

4.2 IoT enabled by 5G and subsequent technologies

New, revolutionary possibilities will be made possible by improvements in wireless technology, which will unlock the potential of Internet of Things technologies. The performance, availability, and connection of IoT devices will be enhanced by 5G and the following networks through the utilisation of THz communication, edge intelligence, and distributed AI. The capabilities of the Industrial Internet of Things (IIoT), which include factory automobiles, industrial robots, and wearable electronics, can be substantially improved with 5G and subsequent network generations. This is something that is currently unattainable. Nevertheless, modern wireless communication-based IoT systems require rigorous security and reliability protocols due to the enhanced openness in their

architecture [101]. Deploying the IoT extensively in critical sectors is exceedingly difficult because of data privacy and security issues [101].

There are both old and new security and privacy issues in the context of the IoT communications environment that 5G and later networks enable. In addition to the primary security concerns highlighted in Sects. 2.1 and 2.2, there are more matters concerning the security of the IoT within the framework of 5G and beyond [102]:

- **How Efficient is the Security Protocol** Devices in 5G and beyond networks that are part of the Internet of Things typically have little computing power, limited storage capacity, and tiny batteries. Therefore, developing efficient security methods that can balance resource usage with protection is a major concern for 5G and beyond-enabled IoT.
- **The Variety of Devices Used by End Users:** Various digital gadgets, each with its own set of capabilities, storage capacity, and communication protocols, are utilised by the IoT that is made possible by 5G and beyond. It is essential to construct heterogeneous security protocols in order to manage a wide range of devices, technologies, and mechanisms.
- **Keeping Private Data Safe from Unauthorized Access and Disclosure:** 5G and beyond-enabled IoT places a premium on protecting users' privacy and identity. Since many individuals are engaging with the IoT through cloud computing and open-source software, data permission is of the utmost importance.
- Another significant challenge with 5G and beyond-IoT applications is ensuring that critical unauthorised infrastructures may only be accessed by authorised users.
- Robust authentication processes are crucial for the IoT to operate consistently in 5G and beyond contexts. Every device will have its own unique identity when a trust execution environment is built on the blockchain, which increases confidence in crucial industries.

5 Enabling technology for 5G-IoT networks

To make the future of connectivity a reality, the 5G-IoT layers must have solid enabling technology drivers. There have been a plethora of recent inventions, ranging from designs that are more energy efficient to technologies that extend the battery life of Internet of Things devices and enable ultra-low-latency communication—both of which are vital for real-time applications such as industrial automation and autonomous vehicles. Strong security protocols and flexible network designs are crucial for 5G networks to support the vast amount of IoT devices that are expected to be connected to them. In addition, by integrating AI, ML, and edge computing, the 5G-IoT framework offers a groundbreaking opportunity to automate various sectors, enhance data analysis, and process data near to its source. If we want 5G-IoT to live up to its revolutionary and life-improving promises, we need to do these things in tandem with appropriate legislative frameworks and get businesses and universities involved.

Integrating wireless and wired network technologies is frequently required when carrying out large-scale IoT projects to meet the future's demands for ultra-connectivity with today's network technology. 5G has the potential to provide the reliability, speed, mobility, scalability, and security that mission-critical facilities within the IoT ecosystem require [102]. There will be a peak of 100% adoption of consumer IoT technologies by the year 2030 [102]. The basic requirements and pervasive connection offered by 5G are

attractive to end consumers. To facilitate a vast array of devices, certain requirements must be met, such as minimal latency, maximum throughput, great adaptability, effective power usage systems, and rapid data transfer. Tens of billions of connections will be able to use the upcoming 5G technology, which will also have a latency of just 1 ms, a bandwidth efficiency of 10 Gbps, and other impressive specs [102].

5G mobile network will expand the amount of IoT applications, such as smart thermostats, kitchen appliances, security cameras, and more [102]. The 5G-enabled Internet of Things incorporates a number of important enabling technologies, such as physical connection and IoT applications. Figure 2 lays out the main enabling technology drivers for 5G-IoT. we can see a compilation of the newest research on the application, architecture, and communication layers, along with an analysis of their pros, cons, and possible remedies. Driving forces behind technology necessitate uniformity.

6 The obstacles, potentials and risks of 5G-IoT

Cybersecurity and resilience for 5G and beyond-enabled IoT; trends, tactics, important patterns, mechanisms, and issues were the research challenges that emerged from our analysis of the included studies in this survey. We also offered some potential solutions to these problems.

1. Emerging Security Flaws: one new security risk in 5G networks is device fingerprinting [102], which arises because 5G's security architecture differs from 4G's. None of the research included in this SLR attempted to fix the recently discovered 5G security holes (such as fingerprinting devices). Consequently, while designing IoT systems that may take advantage of 5G and beyond, we must think about how to handle emerging security threats. One potential solution to this new vulnerability is for network service providers to implement an authentication process. This will allow devices to utilize their radio capabilities once proper security is in place. Attacks that drain power from Narrowband (NB)-IoT devices can be caused by changing specific fundamental network capabilities.
2. Optimal risk prevention approaches are inadequate due to the lack of a generally acknowledged standard architecture for 5G, according to our research on security standards. Cyber risk management strategies for 5G and beyond-enabled IoT have been developed by a variety of academics and service providers.

For IoT systems enabled by 5G and beyond, it will be essential to have a generic benchmark based on any respected standard bodies like NIST, ISO, or IEC. This will serve as the foundation for assessing cyber risk in 5G and beyond-enabled IoT, as well as fostering a shared comprehension of emerging communication systems.

6.1 Communication for self-healing

For the IoT to function reliably and quickly in the face of failures and cyberattacks, self-healing capabilities beyond 5G communication networks are essential. A sufficient level of privacy and security must also be ensured by this. The only study that specifically addressed self-healing was. The absence of deep learning (DL) approaches and real-time scenario performance evaluations are major drawbacks.

One possible solution is to create, test, and refine mutual authentication protocols and autonomous, dynamic adaptive key management methods for use in 5G and future IoT

communication networks. This will help ensure that all of the many nodes, users, servers, and devices involved are legitimate.

6.2 Evaluating real-world problems through empirical research

Concerning the creation of security measures for 5G and beyond-enabled IoT, there is a dearth of practical empirical research that is application-oriented. Research included in this SLR also supports this idea.

One possible solution is to create testbeds for the IoT that are capable of 5G and beyond. These would help with studying and assessing different cybersecurity methodologies, tools, and techniques for IoT that are capable of 5G and beyond.

6.3 Inadequate method for verification

Users may have access to the core network and radio access capabilities in 5G and later Internet of Things systems without complying with the correct authentication procedure. An active or passive attacker can use this vulnerability to steal devices' identities in 5G and future networks. Several researches have been conducted a number of studies [4, 4, 5, 12, 15, 22, 25, 27, 29, 31, 34, 35, 37–39, 53, 54, 103–105] that have concentrated on authentication within the framework of 5G and beyond-enabled IoT. But they have problems with scalability and adaptation, and are generally static.

An interim solution that takes into account the scalability and dynamic nature of the edge and the Internet of Things is to create dynamic authentication procedures that are based on advanced data analytics or fuzzy logic. This will allow us to study the new vulnerabilities in 5G networks.

6.4 The Mannequin test

The likelihood of eavesdropping via the communication channel will be reduced when 5G and subsequent networks tightly handle data flow between end-to-end devices. On the other hand, a major security worry could be man-in-the-middle attacks targeting 5G and other frequencies. You can conclude that many studies dealt with intrusion detection [28, 33, 39, 106]. Nevertheless, man-in-the-middle assaults are not taken into account while evaluating these.

To combat man-in-the-middle attacks, new intrusion detection mechanisms should be created or improved upon, lowering the likelihood of communication-channel eavesdropping in IoT devices enabled by 5G and beyond.

6.5 Ensuring secure communication in environments prone to extreme wave denying

Many use cases and applications, including those in extremely wave-denied environments, will rely on 5G and beyond. To handle such environments, 6G might use molecular communication (MC) devices as a new physical layer. Due to the sensitive nature of the material being transmitted, there are a number of privacy and security concerns surrounding the authentication, encryption, and communication processes involved. Attacks like collision, jamming, and tampering are possible with MC, according to multiple researchers [107–109]. However, our research in this paper shows that few studies have looked at the privacy and security of MC technology.

A possible interim solution is that molecular signals within our bodies can serve as directives for device synchronisation, medication distribution, and safe inter-entity

communication. So that only authorised entities may access the data, it is vital to have an efficient authentication system for 5G and beyond-enabled IoT applications that uses MC technology.

6.6 The safety of AI

In order to guarantee the security and resilience of the IoT in the context of 5G and beyond, several research have used various AI methodologies as the foundational approach, [16, 28, 41, 53, 103, 106, 110]. Unfortunately, AI security has not been adequately addressed in these investigations. When it comes to safeguarding 5G and beyond-enabled Internet of Things applications, artificial intelligence is crucial at different points. Since AI could encounter problems with data resilience, trustworthiness, and ethical concerns, AI security is one of the main obstacles in these systems. Building AI models with the capacity to automatically update their code in response to newly discovered AI security vulnerabilities is one possible alternative.

6.7 Security in internet of things edge computing

Using servers built on the IoT, valuable analytics could be retrieved from a vast array of IoT devices. As can be seen from Uddin et al. [111], this is crucial for many systems, such as 5G and beyond-enabled IoVs/smart traffic. On the other hand, data providers face privacy issues when they allow edge applications to access their embedded services directly.

A possible workaround is to implement privacy-preserving models and procedures in 5G and beyond-enabled IoT. To aggregate data in a way that protects users' privacy, blockchain and federated learning have found widespread application in industries like energy and healthcare; these technologies may hold the key to solving the problem we just discussed.

6.8 Deficit in automated and dynamic trust management

Communication between Internet of Things devices and remote servers is essential in 5G and beyond-enabled IoT systems. There isn't a framework or method that allows automatic trustworthy communication, although there are methods and technologies that can do this.

An exploratory approach that uses artificial intelligence approaches for automated and dynamic trust assessment, prediction, and management is suggested as a potential solution. It is also necessary to provide a lightweight trust management method specifically for Internet of Things systems. In a rapping view, Table 3 presents an overview of the main published research's relevant contributions.

7 Conclusion and future scope

Capable 5G IoT applications and physical connectivity are among the essential enabling technologies that make up the IoT. This article considered a variety of technical standpoints, including security, opportunities, threats associated with 5G-enabled IoT, and the technology factors facilitating its growth. It has been demonstrated that the IoT must be a part of all future applications and services; this calls for a massive infrastructure, a large number of device nodes, and a flexible and wide bandwidth spectrum. The IoT relies on 5G to meet these demands. Even though 5G-enabled IoT can handle the load

Table 3 Overview of existing research contributions

Authors	Proposed approach	Underlying technique
[31]	Basic user authentication scheme	Symmetric key cryptographic approach
[37]	On-the-fly key creation strategy	Entropy-based distinct key generation
[112]	Strategy for securing personal data	Utilizes hash functions and XOR operations
[111]	Lightweight decentralized design for future networks	Blockchain-based digital ledger
[29]	Mechanism to verify remote data integrity	Combines ECC, RSA, and hash algorithms
[103]	Authentication framework for IoT in 5G eHealth	Uses biometric identifiers
[113]	Cybersecurity user identity validation	Integrates neural networks and blockchain
[114]	Leveraging AI and IoT for 5G infrastructure	Uses artificial intelligence and blockchain
[115]	Risk-aware security method	Based on visibility graphs and blockchain
[10]	Design strategy for fog and edge networks	Employs cloud computing technologies
[11]	Healthcare system with remote capabilities	Machine learning with autonomous operation
[12]	Structural design for secure authentication	Utilizes aggregate signature mechanisms
[13]	Countermeasures against jamming threats	Frequency hopping technology
[14]	Distributed risk evaluation system	Based on Software-Defined Networking (SDN)
[15]	Secure user validation protocol	Ensures seamless handovers in mobile systems
[16]	Secure slicing model for 5G networks	Employs neural network models
[116]	Full-scope slice management and control	Adaptive orchestration for 5G environments
[117]	Strategy for system-wide resilience	Built on graph-based dependency models
[54]	Secure cluster-based design	Uses SDN concepts
[22]	Enhanced identity verification model	Applies multiparty computation
[118]	Blockchain-powered EHR system	Digital ledger for decentralized storage
[105]	Privacy-first collision avoidance	Certificate-free signcryption mechanism
[52]	Ultra-light encryption technique	Lightweight cryptographic algorithms
[119]	Robust smart home infrastructure	Based on graph theory principles
[120]	Integrated model for secure networking	Combines SSL, TLS, and SDN protocols
[106]	Surveillance-oriented security system	Powered by artificial intelligence
[110]	Predictive analytics using time data	Machine learning-based forecasting
[23]	E-health platform utilizing 5G	Innovative radio technologies for 5G
[24]	Enhanced network partitioning method	Uses VPN protocols for segmentation
[25]	Lightweight authentication access control	Validated via the AVISPA tool
[26]	Simulated 5G testing environment	Localized clustering-based machines
[27]	Trust-oriented security defense	Incorporates internal/external scenarios
[28]	Intelligent intrusion prevention system	Reinforcement learning approach
[5]	Comprehensive identity verification system	Based on symmetric key encryption
[9]	Protection for multimedia communications	Uses hybrid cipher mechanisms
[38]	Multiple-factor identity verification protocol	Uses symmetric cryptographic design
[32]	Mobile core network with no state tracking	Geographic mobility adaptation
[33]	Detection system for malware threats	Collaborative machine learning
[34]	Secure handover between base stations	Evaluated using AVISPA
[35]	Short-lived secure key generation	Uses probabilistic algorithms
[53]	Authentication using multiple digital agents	Security-based cyber twin framework
[104]	Secure tokenized validation system	Focused on maintaining confidentiality
[4]	Secure identity exchange protocol	Relies on symmetric key methods
[39]	Detection model for botnet activities	Uses DGA, ML, and privacy safeguards
[41]	Autonomous network repair analysis	Based on time series modeling

of future IoT applications, there are several challenges with this new prototype that need fixing, such as scalability, secure communications, and standardization. These problems could be reduced by a universally compatible OS that promotes the development of state-of-the-art open-source protocols enabling easy device-to-device communication.

By working collectively address these problems, the 5G era of ubiquitous Internet of Things connectivity can be fully realized.

7.1 Future scope

The integration of 5G and the Internet of Things has the potential to revolutionise several industries, including healthcare, smart cities, industrial automation, and autonomous systems. The complexity, diversity, and rapid expansion of 5G-enabled IoT networks exacerbate cybersecurity and resilience concerns. To guarantee the continued safe operation of these systems beyond 6G and beyond, more research must focus on three critical areas:

- **Architectures for 6G-IoT with Security by Design** Security protections should be integrated into future IoT systems from the beginning, rather than being added later on. Every layer of a secure-by-design architecture, from the edge to the distant cloud, must include cryptographic protocols, policy enforcement, and identity management. Modular, microservice-based security operations can better respond to evolving threats and network environments [102].
- **Intelligence that is Federated and Decentralised to Protect Privacy** As edge computing and AI continue to merge, it is more important than ever to protect user privacy without sacrificing performance. Even though Federated Learning (FL) is a great way to train models using decentralised data, it may still be hacked through poisoning and inference attacks. Building trustworthy federated learning frameworks should be a top priority for future efforts. These frameworks should include secure aggregation protocols, multi-level differential privacy, and auditability based on blockchain technology [102].
- **Minimalist Cryptographic Tools for Devices with Limited Resources** The growing number of IoT devices, which have limited processing power and battery life, necessitates post-quantum cryptography solutions that are suitable for 5G and 6G environments. Research should also focus on energy-aware security protocols that take into account the device's purpose, the network's design, and the device's importance when determining the level of protection.
- **Automatic Detection of Threats and Mechanisms for Adaptive Defence** Cyber threats are becoming increasingly complex, making static protection ineffective. The demand for autonomous cybersecurity systems with the ability to detect threats in real-time, track their origins, and react properly is growing, as stated by Systems like these should implement reinforcement learning, graph neural networks, and adversarial training [121].
- **Analysis of Word Meanings and Their Contextual Relevance for Security Purposes** The integration of semantic and context-aware communication models is necessary for the improvement of next-generation networks. Researchers should consider security solutions with semantic awareness to identify vulnerabilities, prevent unauthorised access, and enable fine-grained access control in distributed IoT systems. These solutions should take into consideration the data source, underlying intent, and contextual usage [121].
- **Evaluating and Simulating Security with the Use of Digital Twins** Digital twins provide promise for proactive security review since they simulate the operation of actual IoT devices in a virtual environment. In order to simulate assaults, evaluate

defences, and study how systems react to attacks, suggested to develop cybersecurity platforms that utilise digital twins. This is crucial for essential buildings and services, such as healthcare and transportation [121].

- **Trust and Reputation Management Frameworks** For reliable and scalable interoperability among heterogeneous devices, strong trust and reputation models are crucial [122, 123].
- **Regulation Compliance, Standardization, and Collaboration** A common obstacle to security improvement is the discordance between different suppliers, platforms, and standards. It is essential to set up security frameworks that work together, allowing different systems to communicate with each other, following international standards like ISO, NIST, and ETSI, and meeting both national and industry-specific legislation. Questions about policy-driven orchestration of security for several tenants on shared 5G networks will also be asked of interviewees [124].
- **Security Models that Can Withstand Quantum Attacks and the Future** Traditional methods of encryption are becoming more and more useless due to quantum computing. In order to ensure the safety of the IoT, it is necessary to include cryptographic primitives that are resistant to quantum attacks, protocols for securely exchanging keys, and hybrid security stacks that combine classical and quantum protections from the beginning [89].
- **Focusing on Individuals for Security and Usability** There must be no trade-off between usability and security as the number of IoT devices in smart homes, healthcare, and consumer electronics continues to grow. Research in the future should be on creating user-centric cybersecurity systems with an emphasis on clear security laws, user-friendly interfaces, and user-centred threat mitigation measures [89].

Author contributions

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